Robustifying Profile Information Propagation in Profile-Guided Optimization

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Performance is Critical

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- **Profile-Guided Optimization** is a way to achieve great performance
  - Optimization process **tailored** to the workload of the compiled program
  - **Profile** = Execution frequencies of program regions
  - Eliminates the need to infer such information
  - Perfect for high-load applications

Profile-Guided Optimization

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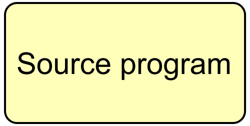

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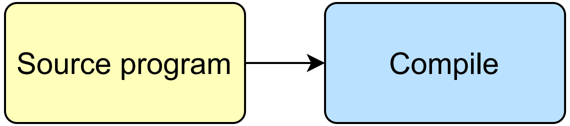

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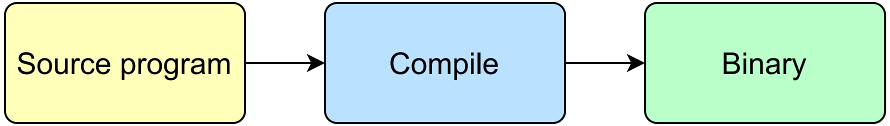

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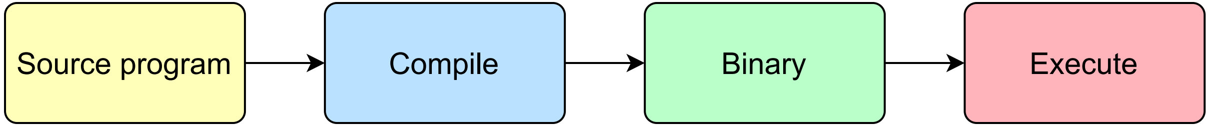

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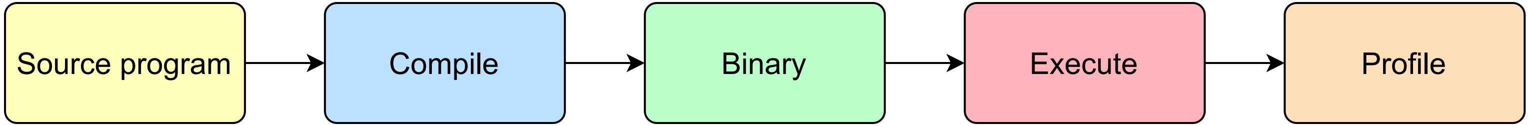

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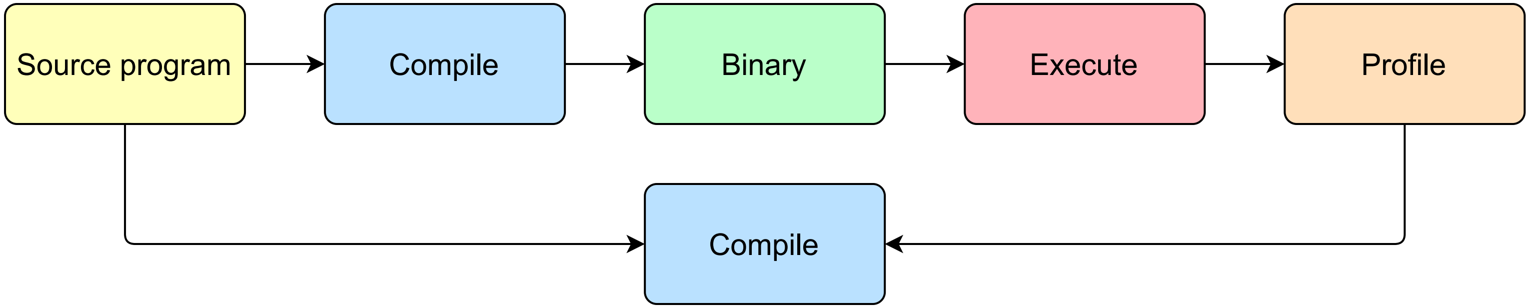

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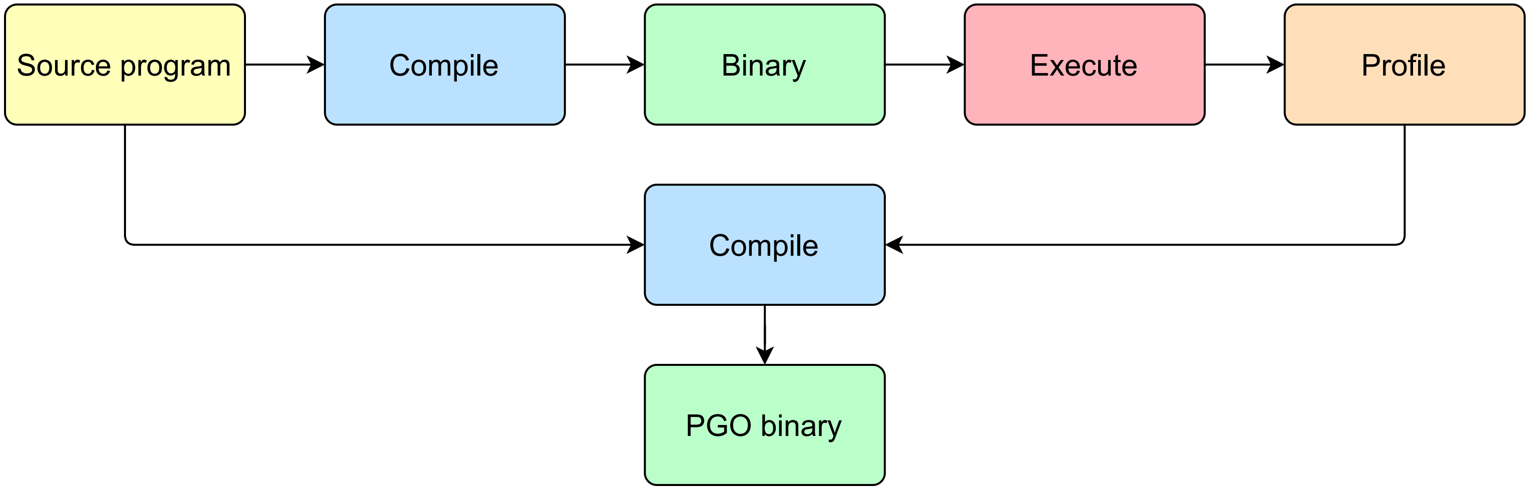

Profile Life-cycle

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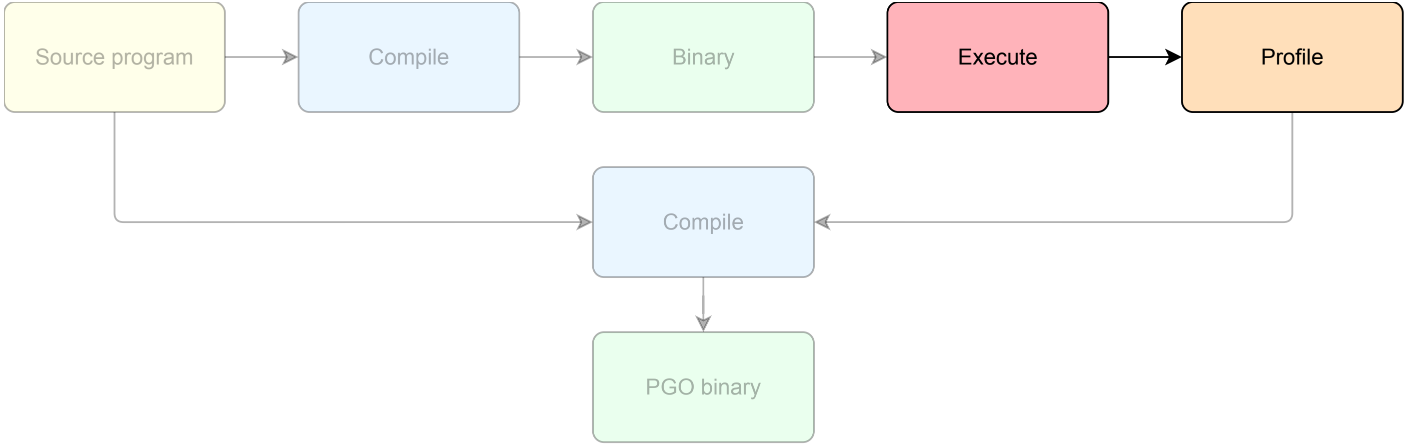

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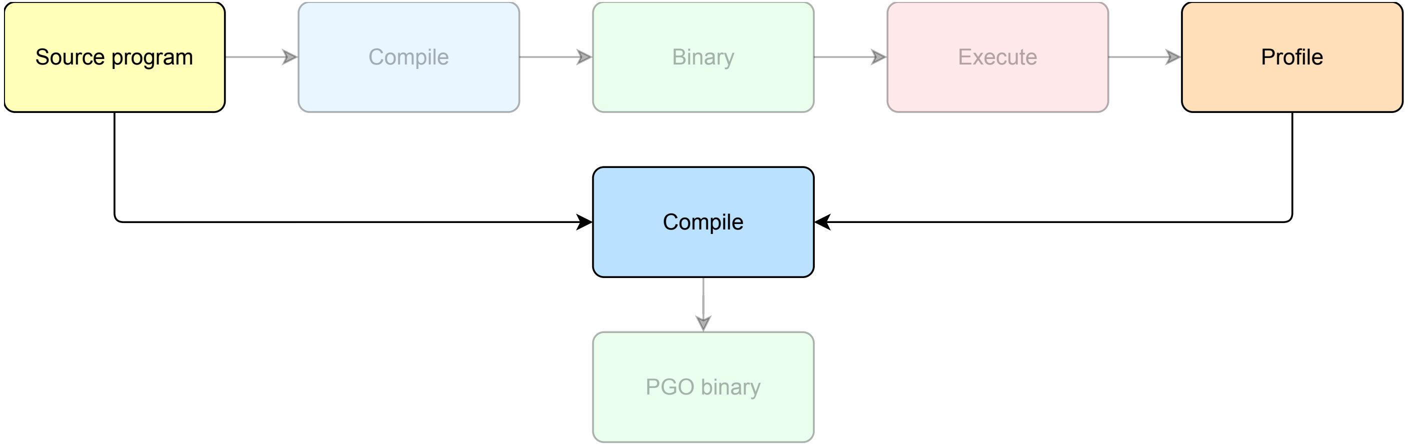

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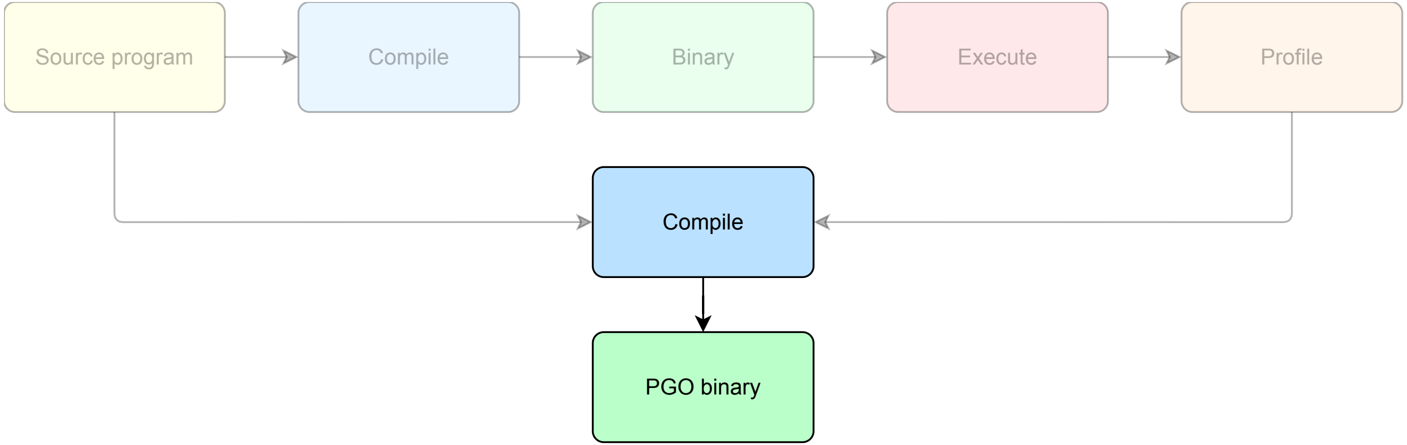
  - **Usage**: The metadata is used to guide optimizing transformations



Practical Obstacles

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- For an ideal PGO application

> Profile must be **complete** and **accurate** in each phase of its life-cycle

- Inaccurate profiles may lead the compiler to make suboptimal optimization decisions.
- In practice, each phase hides sources of inaccuracies
  - **Sampling** strategies are inaccurate by nature
  - The dynamic nature of software leads to **stale profiles**
  - Optimizing passes contain errors in **metadata propagation** logic

State of the Art

1. Wenlei He, Julián Mestre, Sergey Pupyrev, Lei Wang, and Hongtao Yu. “Profile inference revisited”. ↩
2. Amir Ayupov, Maksim Panchenko, and Sergey Pupyrev. “Stale Profile Matching”. ↩
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  - [1] Proposes a rectification algorithm to rectify sampled profiles
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  - [3], [4] only partially address failures in metadata propagation

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Example of Profile Mishandling

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// Before pass
if (x > 0) { // Then branch taken 80 times
    handle_positive(x) 🔥
} else { // Else branch taken 20 times
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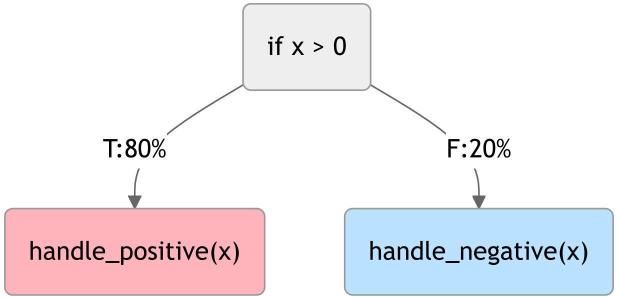

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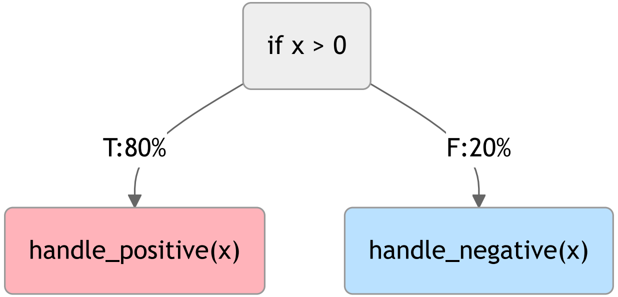

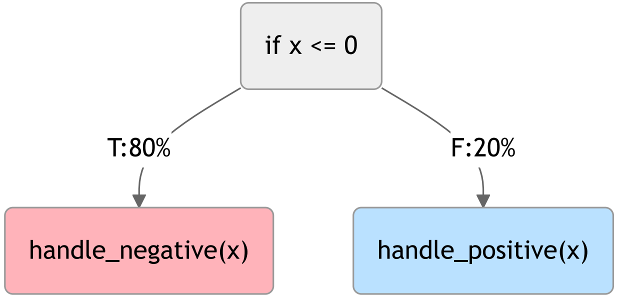

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- **Non-trivial** solution
  - What does it mean to correctly propagate profiles?
  - No immediate correlation between before and after code
  - **Interaction** between passes needs to be taken into account
  - **Static checks** on control-flow are not enough!
  - **No dedicated tools** to validate propagation logic

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> ⚠ Profile information is **transformed** together with the program, but unlike the program itself, its correctness cannot be directly observed.

Research Proposal

Can profile propagation accuracy be assessed systematically?

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  - Stress-test profile propagation logic in order to expose bugs.
  - Off-the-shelf random program generators

Research Proposal

Can profile propagation accuracy be assessed systematically?

- A Methodological and practical framework
- **Black-box** fuzzing strategy
  - Stress-test profile propagation logic in order to expose bugs.
  - Off-the-shelf random program generators
- **Grey-box** fuzzing strategy
  - Lightweight **feedback mechanism** to guide coverage
  - Based on code and **profile** mutations
  - Uses programs from existing **test-suites** as foundation

Black-Box Fuzzing: Methodology

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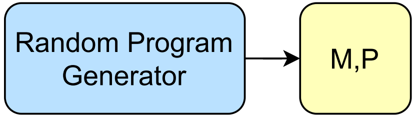

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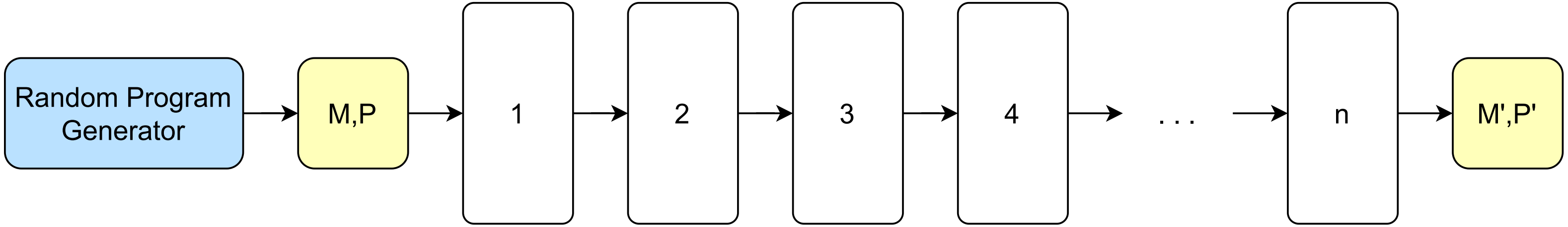

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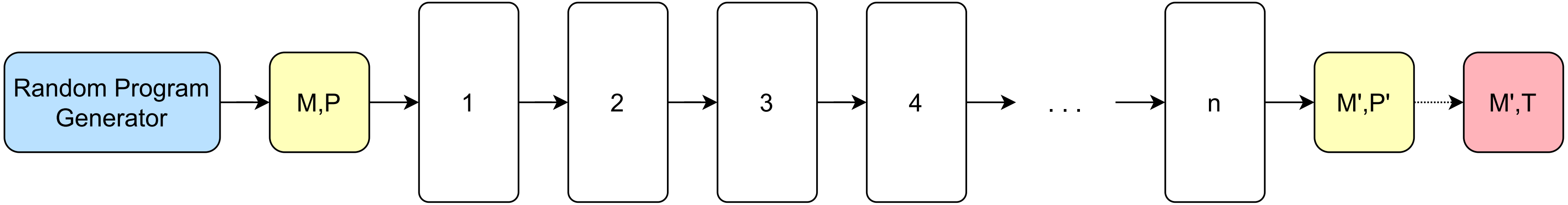

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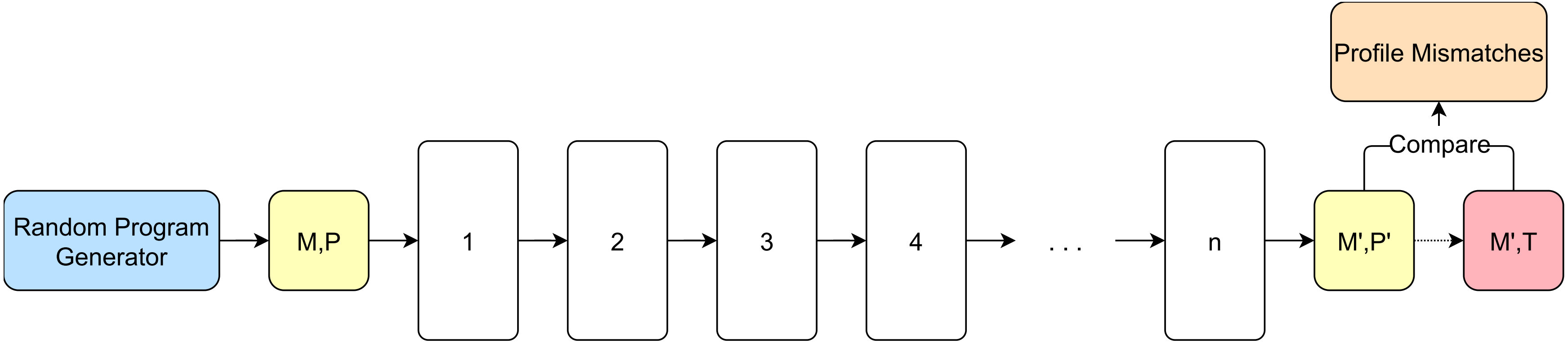
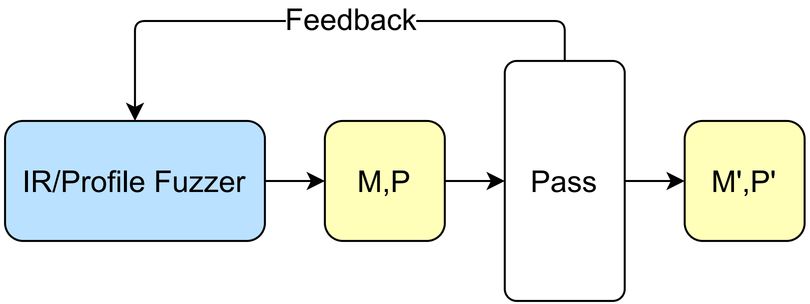


Black-Box Fuzzing: Future Directions

- Improving **automated** bug triaging
  - Classify the large number of found mismatches
  - Avoid reporting multiple instances of the same issue
  - Hard to do due to complexity of randomly generated programs
- Evaluate the framework on **real-world** applications
  - Complete the validation of the methodology
  - SPEC CPU 2017 to evaluate the performance impact of bug fixes

Grey-Box Fuzzing: Methodology



- Mutate both IR and profile metadata
- Instrument optimization passes to obtain profile-aware coverage
  - IR transformations
  - Profile-update actions
- Coverage guides exploration toward new optimization behaviors
- Profile mutations exercise rarely tested PGO scenarios

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Bugs

Performance

Evaluation of the Proposed Directions

- **LLVM** compiler infrastructure as evaluation target
- Can the proposed methodologies:
  - Find new **bugs**? If so, how many?
  - **Improve the performance** of the generated binaries?
  - Improve optimization pass coverage within LLVM?

Bugs | Performance | Coverage

Impacts and Benefits

- **Analysis tools** for compiler developers, to assess their PGO implementations
  - Novel tools to study the problem
  - Enhancement of existing test suites
- **Faster** PGO binaries for the user
  - Reducing the operational costs of deployed applications.
- **Energy savings** due to optimal binaries
  - Reduced environmental footprint.

Thanks for the attention!

Questions?